The sound of silence: Does competition decrease whistleblowing? Evidence from a preregistered experiment*

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April 17, 2026

Abstract

We investigate whether market competition suppresses whistleblowing, using a high-powered, preregistered experiment ($N = 240$). In a setting where firms either operate independently or compete in a tournament for market revenue, managers choose between a productive task and breaking the law to double firm surplus at the expense of the public, while employees decide whether to report misconduct at a personal cost. Aggregate results indicate that competition has a statistically null effect on whistleblowing. While we observe a directional increase in manager lawbreaking under competition, this does not translate into a significant erosion of the propensity to report misconduct. Exploratory analysis reveals heterogeneous treatment effects by population: whistleblowing decreased significantly in our Birmingham pool, but remained unaffected in Cambridge.

Keywords: whistleblowing, market competition, corporate fraud, moral erosion, experiment
JEL Codes: C92, D63, D91, G33, K42, M14

1 Introduction

Market competition is the central organising principle of modern economies, and is traditionally celebrated for driving efficiency and innovation. Classical economic theory posits that competition forces firms to optimise, theoretically driving out inefficient practices, and fostering trust through repeated mutually beneficial trade. However, a persistent counterargument suggests that competition may act as a *corrosive force* on moral behaviour. Shleifer (2004) famously argued that in highly competitive markets, firms may be forced to adopt unethical cost-cutting practices, such as fraud and corruption, simply to survive. In this view, unethical behaviour is not necessarily a product of individual moral failure such as excessive greed, but a structural consequence of the *race to the bottom*. Empirical evidence on this tension remains sharply divided. While some experimental studies find that market interactions significantly erode moral

*Financial support from the Research Priority Area Behavioral Economics of the University of Amsterdam is gratefully acknowledged. The author would like to thank participants at the 13th Oligo Workshop (Cambridge, 2025), the 10th annual conference "Contests: Theory and Evidence" (Reading, 2024), the 2023 ESA World meeting (Lyon, 2023) and the BUE-EBEL 3rd Annual International Conference in Behavioural & Experimental Economics (Cairo, 2023) for their valuable feedback. An earlier version of this paper was included in my PhD dissertation (Ioannidis, 2024).

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concerns compared to individual decision-making (Falk and Szech, 2013; Bartling et al., 2015; Ziegler et al., 2024), others find that competition effects are small and highly context-dependent (Huber et al., 2023).

This ambiguity is particularly critical in the context of corporate fraud. Dyck et al. (2023) estimate that out of all large publicly traded firms, 41% engage in financial accounting frauds and 10% engage in securities fraud, imposing yearly welfare losses of $830 billion in 2021 in the United States alone. Despite its substantial welfare costs and the erosion of public trust in the financial system, only a third of all corporate fraud cases are ever detected (Dyck et al., 2023). Consequently, whistleblowing, defined as the disclosure of illegitimate practices by organisation members, has emerged as a vital tool to expose and deter fraudulent activities (Berger and Lee, 2022; Leder-Luis, 2023). However, the decision to blow the whistle is rarely a simple cost-benefit consideration of legal incentives; it represents a fundamental dilemma, pitting the moral responsibility to prevent social harm against the powerful impulse of loyalty to the firm (Mesmer-Magnus and Viswesvaran, 2005; Waytz et al., 2013; Dungan et al., 2019).

Examining whistleblowing empirically is inherently complex and faces identification and measurement challenges, as only detected fraud and blown whistles can be observed. To address these issues, researchers have increasingly relied on experiments to provide empirical evidence for emerging whistleblowing laws.[1] The experimental literature on whistleblowing has primarily focused on pecuniary factors, such as monetary rewards, protection from dismissal, and the threat of fines (Breuer, 2013; Butler et al., 2020; Mechtenberg et al., 2020; Schmolke and Utikal, 2025). Beyond these incentives, researchers have explored non-pecuniary motivations, including personality traits, diffusion of responsibility, trust in institutions, anticipation of social approval, and social norms (Bartuli et al., 2016; Choo et al., 2019; Antinyan et al., 2020; Butler et al., 2020; Djawadi et al., 2025). A common limitation in these studies, however, is that they focus on the behaviour of experimental firms in isolation. This abstraction ignores the reality that competitive pressure may undermine whistleblowing and facilitate lawbreaking.

Establishing the relationship between competition and whistleblowing is of significant policy relevance. From the perspective of effective regulation, if competition systematically suppresses whistleblowing, it implies that internal monitoring is most fragile in precisely those sectors where market pressure is most intense. Given that interventions to encourage reporting, such as legal protection from retaliation or financial rewards, are costly to implement[2], policymakers must prioritise industries where such measures are most effective. Competitive industries are natural candidates for such prioritisation, as competition may increase managers' incentives to break the law and reduce employees' willingness to report misconduct. Thus, establishing whether such pressure exists is vital to maximise the impact of interventions and more effectively protect public welfare from corporate misconduct.

In this paper, we fill this gap by conducting a high-powered preregistered experiment to test the robustness of whistleblowing under competitive pressure. We compare whistleblowing and

[1] Examples of such laws include the Public Interest Disclosure Act in the United Kingdom (1998), the Dodd-Frank Act in the United States (2010), and the more recent EU Whistleblower Protection Directive in the European Union (2019).

[2] For the EU Whistleblower Protection Directive, European Commission (2018) estimates one-off implementation costs of €746.7 million, and recurring annual costs of €1.2 billion, for public and private sectors combined.

lawbreaking in a setting where firms operate independently to a setting where they compete for market revenue. Our experiment builds upon the whistleblowing game introduced by Butler et al. (2020). In this game, managers choose between a legitimate multiplication task or breaking the law to double firm surplus at the expense of the public. Employees, who benefit from the surplus, must decide whether to report their manager's misconduct at a personal cost. In addition to these primary behavioural choices, we elicit incentivised beliefs regarding the prevalence of such actions, and social appropriateness norms.

Overall, we find little evidence that competition affects unethical behaviour. Our primary results indicate a statistically null effect of competition on whistleblowing. Specifically, we observe an insignificant decrease in whistleblowing under competition, and an insignificant increase in manager lawbreaking. While competition shifted participants' beliefs in the direction of the observed behaviour, it did not significantly affect their underlying social appropriateness norms. These findings suggest that the act of whistleblowing is relatively robust to competitive pressure, which may imply that market intensity need not be the primary concern for regulators when prioritising enforcement resources.

Though exploratory, we document notable heterogeneity in treatment effects across our participant pools. While competition significantly decreased whistleblowing and increased lawbreaking within the University of Birmingham pool, it had no such effect on participants at the University of Cambridge. Taken together, our results suggest that the corrosive effect of competition may be nuanced and population-dependent. We argue that further replication is required to obtain a robust estimate of the effect of competition. Future research could investigate higher levels of competitive intensity, such as winner-takes-all tournaments where only the top firm retains its surplus, or environments featuring survival pressure where failing firms are eliminated, to test the boundary conditions of the null effect reported here.

The remainder of the paper is organised as follows: Section 2 describes the experimental design and theoretical predictions; Section 3 presents the results, distinguishing between preregistered analyses and exploratory findings regarding heterogeneity; and Section 4 provides a concluding discussion.

2 Experimental design and predictions

2.1 The baseline whistleblowing game

The game is based on the whistleblowing game introduced by Butler et al. (2020). Each experimental session involves fifteen participants, who are randomly assigned to one of three roles: manager, employee, or member of the public. Nine participants are organised into three firms, each consisting of one manager and two employees, while the remaining six participants represent members of the public.

The game proceeds in three distinct stages. The first stage is the production stage. The two employees engage in a real-effort addition task, where they are asked to report the sum of six pairs of two-digit numbers. For each correctly solved pair, the employee earns 20 ECU as private earnings and contributes 10 ECU to the firm surplus. Employees are given 120 seconds to complete this stage. This structure ensures that the firm's assets are legitimised by labour, establishing a baseline of moral entitlement before management decisions occur. Members of the public also perform the same task to earn private income, though they do not contribute to any firm surplus.

The second stage is the management stage, which determines the final size of the firm surplus. The manager receives a fixed income of 120 ECU and has the opportunity to double the firm's surplus via one of two options. The first option is doing a multiplication task. The manager is asked to report the product of six pairs of two-digit numbers. If they succeed in computing at least three products correctly, the firm surplus is doubled; otherwise, it remains unchanged. This option represents lawful compliance, but carries execution risk. The second option is breaking the law. The manager skips the multiplication task, and the firm surplus is doubled automatically. However, this choice generates a negative externality, deducting 20 ECU from the earnings of every member of the public. Importantly, the manager does not know the size of the surplus created by the employees when making this decision, which ensures that the choice to break the law is comparable across firms and independent of employee performance.

The third stage is the whistleblowing stage. Employees do not observe the manager's choice directly; instead, we use the strategy method to elicit their conditional willingness to report. Each employee decides whether they are willing to blow the whistle if the manager breaks the law. Reporting comes at a private cost of 25 ECU for the employee and imposes a monetary penalty of 70 ECU on the manager. To mirror the uncertain nature of regulatory sanctions, one of the two employees is randomly selected, and only their decision is implemented. If the manager did not break the law, the employee's decision is irrelevant, and no reporting cost is incurred.

Finally, payoffs are determined by the firm's net surplus. The final surplus is distributed among the firm members: the manager receives 50%, and each employee receives 25%. These profit shares are added to the manager's fixed wage and the employees' private task earnings to determine final compensation.

2.2 The competition treatment

The competition treatment introduces a tournament incentive structure that breaks the independence between firms by linking their final payoffs. This treatment is designed to reflect industries where a firm's revenue depends significantly on its performance relative to its rivals. The first three stages of the game proceed exactly as in the baseline. The modification occurs during the determination of final payoffs. Before redistributing the surpluses to firm members, we rank the net surpluses generated by the three firms within each session. The firm with the largest net surplus is designated the winner, and its surplus is subsequently increased by 50%. The remaining two firms in the session have their surpluses reduced by 25%. In the event of a tie for the largest surplus, the winner is selected randomly with uniform probability. This structure introduces strategic uncertainty, as firm members must form beliefs about the behaviour and task performance of rival firms. Consequently, competitive pressure provides both motives to decrease whistleblowing and increase lawbreaking, as firm members seek to secure the 50% bonus and avoid the 25% penalty.

2.3 Predictions

We specify two predictions regarding the behaviour of managers and employees across the Baseline and Competition treatments. Formal derivations are provided in Appendix A.

The Competition treatment introduces a significant financial incentive to double the firm's surplus, as doing so increases the firm's probability of winning the tournament and securing the largest payoff share. This increase in the expected material benefit of lawbreaking, relative to the multiplication task, provides managers with a strategic incentive to choose the former. This tendency may be further reinforced if competition reduces the perceived moral costs of breaking the law.

Prediction 1. *The propensity of managers to break the law is higher under competition than in the baseline.*

The employee's decision to blow the whistle involves a trade-off between the moral cost of remaining silent and the expected private cost of reporting. Since managers are predicted to break the law more frequently under competition, an employee's decision to report is more likely to be consequential, thereby increasing the expected cost of whistleblowing. If competition also reduces the moral cost of silence, we expect a lower reporting rate.

Prediction 2. *The propensity of employees to blow the whistle is lower under competition than in the baseline.*

Beyond these primary behavioural outcomes, we consider the impact of competition on the underlying psychological drivers of decision-making, namely beliefs and social norms. We expect the competitive tournament to shift normative expectations, leading participants to believe that lawbreaking will be more prevalent and whistleblowing less frequent among their peers. Furthermore, competition may alter the perceived social appropriateness of these actions, potentially attenuating the moral stigma associated with lawbreaking or staying silent. We anticipate that these internal states will correlate with individual behaviour. Specifically, managers who view lawbreaking as more socially appropriate or more common are expected to be more likely to engage in it, while employees who perceive staying silent as more appropriate or more common should be less inclined to report misconduct.

2.4 Implementation

Each session consisted of exactly 15 participants and was played for 12 rounds. Managers retained their role for all rounds, reflecting their lower income variability, while employees and members of the public alternated roles to incorporate the fact that employees in one industry are members of the public for other industries. Employees were randomly assigned to firms in each subsequent round to ensure firms were new. As can be seen from the instructions in Appendix B, our study is a framed experiment, where labels for roles and actions matched the description of the game.[3] To keep decisions comparable and provide cleaner measures of beliefs,

[3] Alekseev et al. (2017) summarised existing evidence and concluded that "using evocative language either does not affect behaviour or affects it in a desirable way by evoking the desired emotional response."

we did not provide feedback on the employees' and managers' decisions between rounds, only on individual performance and firm competition outcomes.

Following the game, participants completed a post-experiment survey. First, we elicit behavioural beliefs, where participants estimated the frequency of whistleblowing and lawbreaking in their session, incentivised for accurate estimates. Next, we elicit social norms of the appropriateness of a manager breaking the law, an employee not blowing the whistle, and the public losing part of their earnings as well as their loyalty to the firm on a Likert scale.[4] These norms were incentivised using the Krupka and Weber (2013) method, rewarding participants for matching the modal response. Finally, we elicit risk preferences using the incentivised Eckel and Grossman (2002) lottery task, and social preferences using the Social Value Orientation scale (Murphy et al., 2011). Standard demographic information was also collected.

The experimental sessions took place between April and October of 2023. Participants were recruited equally from the Birmingham Experimental Economics Laboratory and the Cambridge Experimental and Behavioural Economics Group. The experiment was programmed in oTree (Chen et al., 2016) and preregistered[5]. Ethical approval was obtained prior to data collection.[6] Instructions were provided describing all roles to ensure common knowledge of the game structure. Participants needed to correctly answer comprehension questions before making decisions.

In total we had 240 participants, 120 per treatment arm. Within each treatment arm, we had 60 participants from each participant pool. Each participant took part in only one experimental session. Table 1 breaks down demographic characteristics of participants in all our sessions across treatments and participant pools. Our sessions and treatments are balanced. Participants received an average payment of £11.21 per session for approximately 75 minutes, including a participation fee of £2.00.

	Participant	Age		Gender			Field of study		
Treatment	pool		Male	Female	Other	Social	Natural	Humanities	Other
Baseline	Birmingham	22.4	0.38	0.57	0.05	0.22	0.20	0.10	0.48
	Cambridge	22.4	0.47	0.50	0.03	0.28	0.23	0.18	0.31
Competition	Birmingham	23.1	0.48	0.52	0.00	0.23	0.22	0.13	0.42
	Cambridge	22.5	0.33	0.63	0.04	0.33	0.23	0.25	0.19

Each row corresponds to one combination of treatment and participant pool. Each combination involves 60 participants.

Table 1: Demographics across participant pools for primary treatments

[4]The specific norms we measure are motivated by the management and business ethics literature on rationalisations, which are cognitive strategies employed to reconcile one's unethical actions with moral standards (Ashforth and Anand, 2003; Castro et al., 2020). Relevant rationalisations for whistleblowing are: (i) denial of responsibility, where actors justify their actions believing others in similar positions behave the same way, (ii) selective social comparison, in which individuals see others in different positions as more unethical than themselves, (iii) denial of the victim, perceiving the victim as deserving the harm, and (iv) appeal to higher loyalties, such as firm loyalty.

[5]The preregistration is available at: `https://www.socialscienceregistry.org/trials/11051`.

[6]University of Amsterdam ethical approval reference number: EB-953, University of Birmingham ethical approval reference number: ERN-0894-3, University of Cambridge ethical approval reference number: 23-32.

3 Results

3.1 Preregistered analysis

Figure 1 presents an initial overview of whistleblowing and lawbreaking across our treatments. The figure displays bar graphs for employee whistleblowing (left) and manager lawbreaking (right). To test for treatment effects, we follow our preregistration and econometrically estimate linear probability models (LPM) with standard errors clustered at the session level.[7] Econometric results are presented in Table 2. To ensure the robustness of these findings against potential dependencies within sessions, we also conduct Mann-Whitney rank-sum tests on session-level aggregates. While this aggregation limits us to eight independent observations per treatment and may reduce statistical power, it uses truly independent observations.

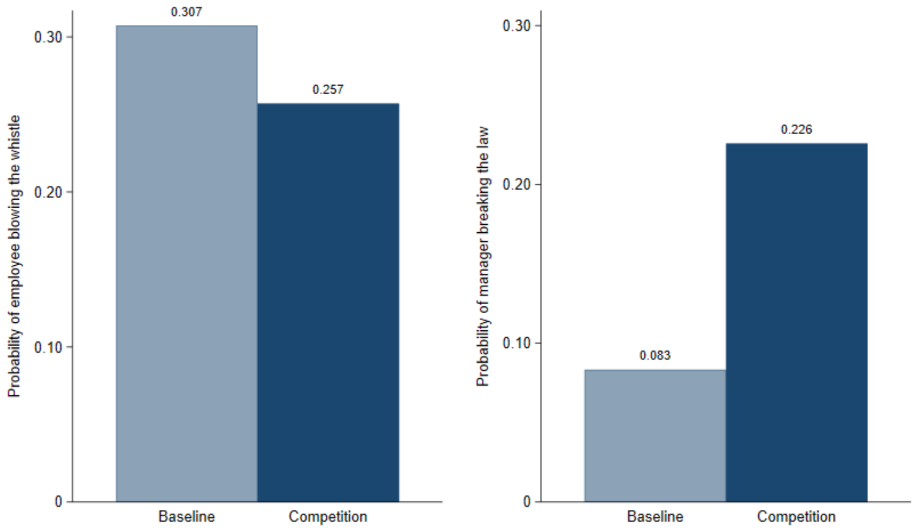

Figure 1: Whistleblowing (left) and lawbreaking (right) across treatments

Regarding whistleblowing behaviour, 30.7% of employees reported misconduct in the Baseline compared to 25.7% in the Competition treatment. Our primary econometric estimation, which controls for risk preferences, social value orientation, and demographics, reveals a negligible and non-significant decrease under competition ($\beta = -0.030, SE = 0.038, p = 0.432$, Table 2, Column 1). The 95% confidence interval for this estimate, $[-0.104, 0.044]$, allows us to rule out reductions in whistleblowing larger than the minimum detectable effect size of 10 percentage points, providing support for the minimal effect of competition on whistleblowing. This result is supported by our non-parametric test (Mann-Whitney rank-sum test, $z = 1.582, p = 0.114, N = 16$).

Result 1. *Competition leads to a non-significant decrease in employee whistleblowing.*

While our primary focus is whistleblowing, we briefly examine manager lawbreaking. In the Baseline, 8.3% of managers chose to break the law, compared to 22.6% under Competition.

[7]The power analysis in our preregistration was based on a LPM. Assuming a baseline whistleblowing probability of 0.23, as in Butler et al. (2020), the minimum detectable effect size with 120 participants per treatment was 0.10. In our baseline, we observe a higher whistleblowing rate (approximately 30%) than in Butler et al. (2020), possibly because our experiment involved repeated rounds whereas the original was a one-shot game.

The primary econometric estimation yields an increase in lawbreaking under competition which, while sitting just outside the traditional significance threshold, is economically non-trivial ($\beta = 0.146, SE = 0.070, p = 0.056$, Table 2, Column 3). Our non-parametric test provides a similar picture, showing a difference that sits exactly on the threshold of traditional significance levels (Mann-Whitney rank-sum test, $z = 1.954, p = 0.051, N = 16$).

Result 2. *Competition leads to a directional and suggestive increase in manager lawbreaking.*

	Blow the whistle		Break the law	
	(1)	(2)	(3)	(4)
Competition	-0.030	0.016	0.146	0.043
	(0.038)	(0.024)	(0.070)	(0.035)
Beliefs				
Whistleblowing		0.707***		-0.170
		(0.046)		(0.134)
Lawbreaking		-0.306**		0.705**
		(0.099)		(0.222)
Social norms				
Employee silence		-0.238		
		(0.124)		
Manager lawbreaking				-0.089
				(0.155)
Public losing earnings		-0.140		0.401*
		(0.075)		(0.198)
Firm loyalty		-0.130*		-0.108
		(0.051)		(0.109)
Controls	Yes	Yes	Yes	Yes
Observations	1,152	1,152	576	576

Standard errors in parentheses, clustered on matching group level.
Significance levels * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Controls: Age, Gender, Study, Risk, Social Value Orientation

Table 2: The effect of beliefs and social norms on whistleblowing and lawbreaking behaviour

Additionally, we examine how beliefs and social norms correlate with behaviour.[8] Whistleblowing is strongly positively correlated with beliefs about peer reporting and negatively correlated with the expected frequency of lawbreaking (Table 2, Column 2). Similarly, managers are significantly more likely to break the law when they perceive such misconduct as prevalent among their peers (Table 2, Column 4). We find little evidence that social norms correlate with behaviour; employees blow the whistle less when they feel more loyal to their firm, and managers break the law more when they find the public losing earnings more acceptable. Overall, decisions are primarily correlated with expectations rather than social norms.

Observation 1. *Whistleblowing and lawbreaking are significantly correlated by beliefs regarding*

[8]Each employee accounted for 6 out of 72 whistleblowing decisions in their session. Similarly, each managers accounted for 12 out of 36 of the lawbreaking decisions. Thus, our belief variable may produce biased results as it is endogenous. Repeating the analysis using modified beliefs, i.e., the originally stated beliefs after subtracting the decisions of each participant, provides qualitatively similar results.

the prevalence of such behaviours, whereas individual social norms have a minimal impact on decisions.

We further investigate whether the competition treatment directly altered beliefs and social norms. Table 3 presents the comparison of standardised beliefs and appropriateness ratings across treatments. We observe that competition successfully shifted participants' expectations in the direction of the observed behaviour. In contrast, social norms remained remarkably stable under competition.

	Belief about frequency of whistleblowing	Belief about frequency of lawbreaking	Appropriateness of an employee staying silent	Appropriateness of a manager breaking the law	Appropriateness of the public losing earnings	Loyalty to the firm
Baseline	0.3539	0.1968	0.4688	0.3125	0.1646	0.5896
Competition	0.2804	0.3148	0.4687	0.2396	0.2042	0.6167
p-value	0.0254	0.0787	0.9045	0.1734	0.1504	0.4984
Observations	192	48	192	48	240	240

All beliefs and social norms are standardised to be between 0 and 1. The p-values are from Mann-Whitney ranksum tests.

Table 3: Beliefs and social norms across treatments

Observation 2. *There is suggestive evidence that competition affected the beliefs of employees and managers, whereas social norms were unaffected by competition.*

3.2 Exploratory heterogeneity analysis

While our primary analysis shows limited treatment effects, we document a stark discrepancy in behaviour between the University of Birmingham and University of Cambridge participant pools. Within the Birmingham sample, competition had a statistically significant impact on behaviour: whistleblowing decreased from 31.9% to 19.1% (ranksum test, $p = 0.021, N = 8$), while manager lawbreaking rose sharply from 0.7% to 18.8% (ranksum test, $p = 0.025, N = 8$). Conversely, for participants at the University of Cambridge, competition affected neither whistleblowing ($p = 0.770$) nor lawbreaking ($p = 0.559$). However, as these population-specific comparisons were not part of our preregistered hypotheses and rely on smaller subsamples, they may be underpowered and should be interpreted cautiously as purely exploratory.

Observation 3. *Within each participant pool, we find that:*

(a) competition significantly decreased whistleblowing and increased lawbreaking within university of Birmingham participants;

(b) competition did not affect whistleblowing or lawbreaking within university of Cambridge participants.

4 Concluding discussion

Across two participant pools, we find that whistleblowing behaviour is not systematically suppressed by competition. From a theoretical standpoint, this result is consistent with our observation that social norms regarding the appropriateness of silence remained remarkably stable.

In the framework of our model, since the moral cost of silence remained constant and beliefs regarding lawbreaking did not shift sufficiently to alter the expected cost-benefit trade-off, the decision to blow the whistle remained unchanged.

If competition had systematically suppressed whistleblowing, it would suggest that interventions should be prioritised in highly competitive sectors where internal monitoring is most fragile. Our findings, however, suggest that competition intensity may not be a primary factor for policymakers when deciding which industries to prioritise for whistleblowing interventions. This aligns with recent theoretical work, such as Thanassoulis (2023), which suggests that more competitive markets are not inherently more prone to misconduct.

The importance of these results is underscored by the high statistical power of our design. With 240 participants and 80% power to detect a 10 percentage point reduction in whistleblowing, our findings offer a null result that is often missing due to publication bias. Moreover, our study functions as a small-scale self-replication across two distinct populations, which allows us to gauge boundary conditions for the effect of competition. This approach reveals that an aggregate null result can mask significant underlying heterogeneity, demonstrating that what might appear as a "failed" treatment in a combined sample may actually be a nuanced reflection of context-dependent behaviour.

Documenting heterogeneous effects, such as the discrepancy between the Birmingham and Cambridge pools is essential for the scientific process, and ties into the academic discussion of generalisability. As a thought experiment, consider two researchers running replications of our treatments: one using the Birmingham pool might conclude that competition suppresses whistleblowing, while another using the Cambridge pool would conclude it has no effect. Neither researcher would be strictly incorrect, yet our study illustrates how even experiments can yield divergent conclusions whose causes may be driven by unobserved differences, and are difficult to predict beforehand. Ultimately, our results suggest that the "corrosive" influence of competition is not an inherent economic law, but rather conditional on the population in question. Further research is required to establish the specific conditions under which this effect emerges, prioritising diverse populations to move beyond the limitations of single-pool observations.

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Appendix A Formal derivations of experimental predictions

We denote the probability with which the manager breaks the law in treatment j by b^j, and the probability with which the selected employee blows the whistle in treatment j by w^j, where $j \in \{B, C\}$ indicates the baseline and competition treatments respectively. We also denote by d_b^j and d_s^j respectively, the moral cost of the manager for breaking the law and the moral cost of the employee for not blowing the whistle in treatment j. Finally, we denote the expected firm surplus produced by the two employees before the manager makes any decision by S, and the probability that the manager can correctly solve the six multiplication problems by a.

A.1 Manager utility and lawbreaking comparative statics

In the Baseline treatment, breaking the law yields the manager their fixed income and half of the doubled surplus (S), while reducing utility by the moral cost of the act and the expected punishment.

$$U_b^B = 120 + S - 70w^B - d_b^B$$

Choosing the multiplication task yields the fixed income and half of the expected surplus, which is doubled only if the task is successfully completed.

$$U_t^B = 120 + \frac{1}{2}\left[a \times 2S + (1-a) \times S\right] = 120 + \frac{1+a}{2}S$$

The manager weakly prefers to break the law if $U_b^B \geq U_t^B$, which implies:

$$\frac{1-a}{2}S \geq 70w^B + d_b^B \tag{1}$$

Intuitively, a manager is more likely to break the law if they are of low ability, if whistleblowing is less likely, or if their moral costs of doing so are low.

To derive the condition for the Competition treatment, we let π_{1S} denote the probability of winning the competition if the firm surplus is not doubled, and π_{2S} if it is doubled. By definition, $\pi_{2S} \geq \pi_{1S}$. Given that ties are broken randomly and initial surplus is expected to be uniform across firms[9], π_{2S} is bounded below by 1/3 (the case where all firms double their surplus) and π_{1S} is bounded above by 1/3 (the case where no firms double their surplus). Thus, $\pi_{2S} \geq \frac{1}{3} \geq \pi_{1S}$.

In the Competition treatment, the utility from breaking the law is (after rearranging terms):

$$120 + \frac{3(1+\pi_{2S})}{4}S - 70w^C - d_b^C$$

The utility from the multiplication task is:

$$120 + \left[\frac{3a(1+\pi_{2S})}{4} + \frac{3(1-a)(1+\pi_{1S})}{8}\right]S$$

The manager prefers to break the law if $U_b^C \geq U_t^C$, which simplifies to:

$$\frac{1-a}{2}\left[\frac{3}{2}\left(\frac{1}{2} + \pi_{2S} - \frac{\pi_{1S}}{2}\right)\right]S \geq 70w^C + d_b^C \tag{2}$$

Since $\pi_{2S} \geq \frac{1}{3} \geq \pi_{1S}$, the term in the square brackets on the left-hand side is weakly larger than 1. This indicates that the expected material benefit of lawbreaking is higher under Competition than in the Baseline. Assuming $w^C \leq w^B$ and $d_b^C \leq d_b^B$, managers are predicted to break the law more frequently under competition.

[9] The addition task is straightforward and almost all employees are expected to produce the maximal surplus. Out of 2,016 sets of the six addition problems that the employees in our experiment worked on, they solved all six correctly in 1,888 of them (93.6%) with almost all other instances resulting in five correct. Thus, the surplus of each firm before the manager decision was roughly fixed at 120, 60 from each employee.

A.2 Employee utility and whistleblowing comparative statics

For an employee, the expected cost of blowing the whistle is $0.5 \times 25 \times b^j$, as the cost is only incurred if the manager chose to break the law and the employee's decision is randomly selected for implementation. The benefit is the avoidance of the moral cost d_s^j. In the Baseline, the employee prefers to report if:

$$d_s^B \geq \frac{25}{2} b^B \tag{3}$$

In the Competition treatment, the condition is:

$$d_s^C \geq \frac{25}{2} b^C \tag{4}$$

Given Prediction 1 ($b^C \geq b^B$), the expected cost of reporting is higher under competition. Consequently, we predict that employees will be less likely to blow the whistle in the Competition treatment, a tendency that would be further amplified if competition also reduces the moral cost of silence ($d_s^C \leq d_s^B$).

Appendix B Instructions

B.1 Welcome screen

Welcome

Welcome to this experiment. Please read the following instructions carefully. We ask that you do not communicate with other participants during the experiment. The use of mobile phones is not allowed during this experiment. If you have any questions, or need assistance of any kind, at any time, an experimenter will assist you privately. The data collected through this experiment does not include your name or any other information that would allow your identification. All the data you provide during the experiment cannot be traced back to you.

Payment

In addition to your participation fee, you may earn substantially more money from today's experiment. You will be paid privately and anonymously in cash at the end of the experimental session today. Earnings during the experiment will be denominated in Experimental Currency Units, or ECU. Each ECU is worth £0.01. The participation fee is £2.00. After the experiment finishes, you will be paid the money you earned plus your participation fee.

Duration

You will be asked to make decisions in 12 rounds.

B.2 Tasks and decisions

Employee Task

The two employees have the task of adding two numbers and report their sum. Each correct answer gives them 20 points. Additionally each correct answer by each employee generates 10 points to the firm surplus. In total, each employee will be given 6 pairs of numbers to add. Employees have 120 seconds to solve as many as they can.

Manager Decision & Task

Managers have the opportunity to double the firm surplus. Their decision is to choose how they want to try to double the surplus. They can do so in two ways:

- Do the Manager Task: In the task, managers are asked to multiply two numbers and report their product. Managers will be given 6 pairs of numbers to multiply. Managers have 120 seconds to solve as many as they can. If they give at least 3 correct answers, the firm surplus is doubled.
- Break the law: If the manager breaks the law, they skip the Manager Task and the firm surplus will be doubled automatically. However, it will also generate a loss of 20 points to each of the 6 members of the public.

Employee Decision

The employees have the option to blow the whistle if their manager broke the law. The decision to blow the whistle will be relevant ONLY if the manager broke the law. If the manager did not break the law, this decision will not be implemented. The employees will make this decision before the manager makes their decision. Each employee can decide whether they are willing to blow the whistle. Blowing the whistle will cost the selected employee 25 points, and will generate a penalty of 70 points to the manager. This penalty will be removed from the manager's earnings. One of the two employees will randomly be selected and their decision will be implemented. The decision of the other employee will not be implemented.

Members of the Public Task

The members of the public have the task of adding two numbers and report their sum. Each correct answer gives them 20 points. In total, each member of the public will be given 6 pairs of numbers to add. Members of the public have 120 seconds to solve as many as they can.

B.3 Payoff explanations

Winning firm [Competition treatment]

One of the three firms will win the competition and their surplus will be further increased by 50%. Each of the other two firms that did not win the competition will have their surplus reduced by 25%. The winning firm will be the firm with the largest surplus.

Surplus distribution

The total firm surplus will be distributed as follows: The manager keeps 50% of the surplus, and each employee will keep 25%.

Payoff explanation

- Employees:
  - Earn 20 points for each correct answer in number-adding task.
  - Earn 25% of the final firm surplus.
  - Lose 25 points if they blow the whistle and their decision is implemented.
- Managers:
  - Earn fixed income of 120 points.
  - Earn 50% of the final firm surplus.
  - Lose 70 points if they broke the law and the selected employee blew the whistle.
- Members of the public:
  - Earn 20 points for each correct answer in number-adding task. [Task treatments]
  - Lose 20 points for each manager that broke the law.

Paid rounds

At the end of the session, 4 rounds will be randomly selected for payment.

B.4 Survey

Estimates

Please answer the following questions. For each question, you will receive additional 20 points if your answer is within a range of 3 of the actual number. Your earnings from this task will be added to your earnings from the game.

- The managers had the choice between breaking the law and doing the multiplication task 36 times in total during the experiment. How many times do you think they broke the law?
- The employees had the choice to blow the whistle or not 72 times in total during the experiment. How many times do you think they blew the whistle?

Social norms

Please answer the following questions. For each question, you will receive additional 10 points if your answer matches the answer of the majority of the participants in today's experiment. Your earnings from this task will be added to your earnings from the game.

- How would you judge a manager who broke the law?
  (Very immoral, Immoral, No judgement, Moral, Very moral)

- How would you judge an employee who did not blow the whistle?
  (Very immoral, Immoral, No judgement, Moral, Very moral)

- How do you feel about the members of the public losing earnings because managers broke the law?
  (Very unacceptable, Unacceptable, No judgement, Acceptable, Very acceptable)

- In the rounds that you were part of a firm, how loyal did you feel to the firm?
  (Not loyal at all, Not very loyal, Neutral, Loyal, Very loyal)

Lottery task

In the following task, 6 different lotteries are presented on your screen. In each of these lotteries, both rewards A and B are equally likely, i.e. have a probability of exactly 50%. The rewards are denoted in points.

You are asked to choose exactly one of the lotteries, which subsequently will be implemented. A random generator will determine whether you win reward A or reward B, respectively. At the end of the experiment, your reward will be added to you earnings.

	Lottery 1	Lottery 2	Lottery 3	Lottery 4	Lottery 5
Reward A (50%)	140	120	100	80	60
Reward B (50%)	140	180	220	260	300

Distribution task

In this task you have been randomly paired with another person, whom we will refer to as the other. This other person is someone you do not know and will remain mutually anonymous. All of your choices are completely confidential. You will be making a series of 6 decisions about allocating resources between you and this other person. For each of the following questions, please indicate the distribution you prefer most by choosing the button along the midline. You can only choose one distribution for each of the 6 questions. Your decisions will yield money for both yourself and the other person.

There are no right or wrong answers, this is all about personal preferences. One of the 6 decisions will randomly be selected and implemented. At the end of the experiment, the outcome will be added to you earnings.

Option	1	2	3	4	5	6	7	8	9
You receive	85	85	85	85	85	85	85	85	85
Other receives	85	76	68	59	50	41	33	24	15

Option	1	2	3	4	5	6	7	8	9
You receive	85	87	89	91	93	94	96	98	100
Other receives	15	19	24	28	33	37	41	46	50

Option	1	2	3	4	5	6	7	8	9
You receive	50	54	58	63	68	72	76	81	85
Other receives	100	98	96	94	93	91	89	87	85

Option	1	2	3	4	5	6	7	8	9
You receive	50	54	59	63	68	72	76	81	85
Other receives	100	89	79	68	58	47	36	26	15

Option	1	2	3	4	5	6	7	8	9
You receive	100	94	88	81	75	69	63	56	50
Other receives	50	56	63	69	75	81	88	94	100

Option	1	2	3	4	5	6	7	8	9
You receive	100	98	96	94	93	91	89	87	85
Other receives	50	54	59	63	68	72	76	81	85

Demographics

Please enter the following information.

- Please indicate your age.
- Please indicate your field of study.
  (Economics, Social Sciences, Natural Sciences, Humanities, Applied Sciences, Other)
- Please indicate your gender.
  (Male, Female, Prefer not to answer)